ICCCI 2026 - Response to Reviewers

MeReader: An Offline, Privacy-Preserving, Progress-Aware AI Assistant for Narrative eBook Reading (submission 33)

We thank the reviewers for their time and constructive feedback. Each comment and review has been considered and the manuscript has been revised accordingly. Below, we address each review individually.

Review 1 - "While the models' accuracy and performance are clearly discussed, further clarification on how they are integrated into the MeReader system would be helpful. For instance, whether they were fine-tuned specifically for this reading task or if general-purpose models were adapted, and how this affects performance in specific reading contexts. The paper presents an interesting and practical solution for enhancing narrative eBook reading while accounting for privacy considerations. I am inclined to accept it, with the suggestion to include more details on the model integration and fine-tuning process to fully showcase the system's capabilities."

Response: MeReader uses pre-trained, off-the-shelf local models through Ollama. None are fine-tuned. This is a deliberate design choice, because fine-tuning on narrative text risks memorisation and content leakage, as shown by Mireshghallah et al.[1] and Ippolito et al.[2]. Instead of baking book knowledge into model weights, the system supplies context at query time through retrieval-augmented prompting. The same local LLM handles answer generation, query expansion, and optional passage reranking, each with a different temperature. The embedding model is nomic-embed-text, producing 768-dimensional vectors stored in Qdrant. All models are configurable via environment variables, which is how the cross-model evaluation in **Section 5** was conducted.

Manuscript changes:

In **Section 4**:

> *MeReader uses pre-trained, off-the-shelf local language models via Ollama, with no specific fine-tuning. This is a deliberate design choice because fine-tuning on narrative content would risk memorization and content leakage, consistent with findings by Mireshghallah et al. [19] and Ippolito et al. [11]. Instead, the system relies entirely on retrieval-augmented prompting for context. The LLM serves three roles within the pipeline: (i) answer generation (temperature 0.3 for evaluation, 0.7 for interactive use), (ii) query expansion for retrieval augmentation (temperature 0.3, generating two alternative queries per item), and (ii) passage re-ranking for complex queries exceeding eight results (temperature 0.1, scoring on a 1–10 scale). The system prompt enforces several constraints, including excerpt-only answering, spoiler prevention, and a fallback response when context is insufficient. Embeddings are generated using nomic-embed-text, producing 768-dimensional vectors stored in Qdrant. All models are configurable via environment variables, enabling the evaluation reported in Section 5.*

We also expanded the storage layer description:

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[1] F. Mireshghallah, A. Uniyal, T. Wang, D. Evans, T. Berg-Kirkpatrick, "An Empirical Analysis of Memorization in Fine-tuned Autoregressive Language Models," in Proc. EMNLP 2022, pp. 1816–1826. doi:10.18653/v1/2022.emnlp-main.119

[2] D. Ippolito et al., "Preventing Verbatim Memorization in Language Models Gives a False Sense of Privacy," arXiv:2210.17546, 2023. doi:10.48550/arXiv.2210.17546

The storage layer employs a two-tier database strategy. SQLite (via SQLAlchemy) for structured metadata and Qdrant for vector embeddings. The SQLite schema uses UUIDs as primary keys with foreign keys maintaining referential integrity between Book, Chapter, and ReadingProgress entities, with cascade deletion on book removal.

And we clarified how the progress boundary works during retrieval:

The progress boundary is calculated as min(total _positions, max(1, current_position)), handling edge cases at the beginning and end of a book. This boundary is enforced at the Qdrant retrieval layer via a Range(lte=boundary) filter, ensuring that future content is excluded before the LLM processes the query.

Review 2 - "Its main limitation is that the novelty lies more in system integration than in a fundamentally new method. Most of the key components are established techniques, and the progress-aware boundary mechanism, while useful, is relatively simple. The evaluation is also limited in scope.
The writing, however, still needs refinement. Grammatical errors, like on page 1, "privacy-perserving" should be "privacy-preserving", on page 10, "Future iterations will incorporate multi-genre books, literature, and long form novels across languages" is clumsy. "books" and "literature" overlap, awkward phrasing, repetition, and uneven style appear throughout the paper, with the introduction and conclusion showing these issues most clearly. In several places, sentences stretch further than the idea requires, and the result is less precision rather than more depth. A careful language revision would make the paper easier to read and would also give the underlying contribution a more polished and convincing presentation."

Response: On novelty, we agree. The individual components are established. The revised manuscript now states the contribution as the synthesis of progress-aware retrieval, local inference, and narrative-specific assistance combined within a single offline reader. The boundary is enforced at the retrieval layer rather than at the end of the pipeline. Three specific design contributions are listed in the **Introduction**.

We corrected the "privacy-perserving" typo. We removed the clumsy sentence about "multi-genre books, literature, and long form novels" and replaced it with a cleaner future work discussion in the **Conclusion**.

We also acknowledge the limitations in the **Conclusion**. The three novels, reference answers, LLM-based judging, and absence of a user study are all stated as boundaries. The future work section now lists broader testing, multilingual and non-fiction books, and human audits as explicit next steps.

Manuscript changes:

The **Introduction** now states the contributions clearly:

[...]. MeReader addresses these shortcomings by implementing the spoiler safety of Kindle [24] into an open architecture (one that is auditable) while also maintaining the deep retrieval memory lacking in the KOReader plugin [8].
This study makes three contributions: (i) a progress-aware retrieval boundary enforced at the vector storage layer, (ii) a hybrid retrieval fusion (BM25, dense vector, summary embeddings,

query expansion) optimized for eBooks (a setting under-explored in RAG literature), and (iii) it evaluates ten local LLMs on comprehension and spoiler-prevention tasks, providing empirical evidence about quality–latency trade-offs on consumer hardware.

The **Related Work** summary now compares MeReader directly with existing systems:

In summary, existing research has explored private AI-assisted reading, semantic retrieval, and RAG independently. However, no existing system combines these properties within a single application for narrative eBook reading. Kindle's proprietary progress gating [24] operates as a closed feature, whereas MeReader implements progress awareness at the vector retrieval layer, making it auditable and extensible. Compared to KOReader's trailing window plugin [8], which loses earlier context when in excess of ~100k characters, MeReader maintains the retrieval memory of the entire book while also enforcing spoiler boundaries through location-tagged filtering. And lastly, unlike general-purpose RAG systems that operate across document collections, MeReader is optimized for single-document eBooks, where retrieval must respect temporal story progression.

The abstract now positions the work without overclaiming:

None of these individual components are new on their own, but no existing system has brought them together in one place. MeReader addresses this gap by combining these capabilities into an offline, constrained assistant that enhances comprehension without compromising continuity or user privacy.

The Conclusion now has a more grounded future work discussion:

For future work, broader testing on additional books, genres, and languages would help determine whether the same retrieval and spoiler-prevention behavior holds under different narrative structures. Specifically, multilingual and non-fiction books are useful targets because they may stress the embedding and retrieval pipeline differently from English fiction. Also, more diverse genres may require different BM25/vector fusion settings, since the balance between exact term and semantic matching can vary with vocabulary and writing style. Furthermore, a human audit could be added alongside the LLM-based judge to provide an independent check on semantic correctness and spoiler safety, since automated judges can still miss subtle errors or over-reward surface similarity.

Review 3 - "… at the same time, the paper would benefit from minor revisions, particularly in further clarifying some experimental design choices and in broadening the discussion of generalizability beyond the current evaluation setting. Overall, the work is sound, relevant, and suitable for acceptance after minor revision."

Response: We revised Section 5 to explain the experimental design choices more explicitly. On book selection, the three novels were picked because they are widely available through online stores (Amazon, Barnes & Noble, Kobo), have clear plot progression with spoiler-sensitive events, and are short enough to evaluate on consumer hardware. BERTScore was chosen over exact match because narrative answers often paraphrase. Exact match would penalize semantically correct responses just because they use different words. Regarding temperature, the evaluation used 0.3 to reduce output variance and make model comparisons cleaner. We added a discussion of generalizability to the **Conclusion**. Rather than claiming the system transfers unchanged to other settings, we state candidly that broader testing across genres, languages, and hardware is needed, and that different text types may require different retrieval weight configurations.

Manuscript changes:

Book selection and benchmark description:

> *We evaluated 600 queries across three novels: 1984 (George Orwell) [9], Of Mice and Men (John Steinbeck) [12], and The Death of Ivan Ilyich (Leo Tolstoy) [16]. These novels were selected because they are widely available in online stores such as Amazon [1], Barnes & Noble [2] and Kobo [27], and they are of moderate length with clear plot progression, making them suitable for retrieval evaluation.*
>
> *The benchmark includes 300 comprehension queries and 300 spoiler-prevention scenarios with their example items shown in Code 1 and Code 2 respectively. Ground-truth answers were generated with Gemini 2.5 Pro. For each item, passages were retrieved using lexical-semantic search, deduplicated, and limited to six excerpts. The models were instructed to answer only from the retrieved excerpts and to respond "Info not available in text so far" when needed. During evaluation, the temperature was set to 0.3 to reduce variance across model comparisons.*

BERTScore justification:

> *BERTScore was chosen over exact match because the model-generated answers are narrative and often paraphrased; exact match would unfairly penalize semantically correct but lexically different responses.*

Conclusion:

> *For future work, broader testing on additional books, genres, and languages would help determine whether the same retrieval and spoiler-prevention behavior holds under different narrative structures. Specifically, multilingual and non-fiction books are useful targets because they may stress the embedding and retrieval pipeline differently from English fiction. Also, more diverse genres may require different BM25/vector fusion settings, since the balance between exact term and semantic matching can vary with vocabulary and writing style. Furthermore, a human audit could be added alongside the LLM-based judge to provide an independent check on semantic correctness and spoiler safety, since automated judges can still miss subtle errors or over-reward surface similarity.*

To end, we believe these revisions address all the concerns that were raised. We once again thank all stakeholders involved for their invaluable feedback.